

Trainings ▾

Remove

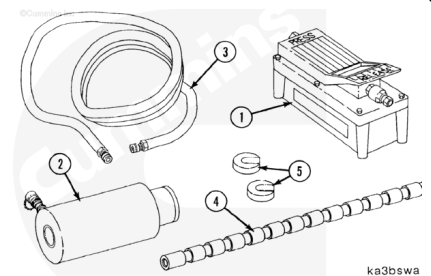
The K38 and K50 engines contain two camshaft bushings in each bore, except in the bores in the left bank (LB) front and right bank (RB) rear. The LB front and RB rear contain one bushing **only**.

When the engines are manufactured, a bushing, Part Number 3175805, which does **not** contain an internal oil groove, is installed in all bores except the LB front and RB rear. To reduce confusion in the field, **only** the bushing, Part Number 205230, with an internal oil groove, is available for service.

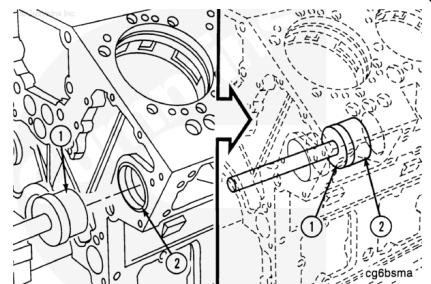
If a bushing in a bore that contains two bushings **must** be replaced, both bushings **must** be replaced. It is recommended that if one bushing of the engine bank **must** be replaced, that all of the bushings for that bank be replaced.

Use the camshaft bushing driver assembly, Part Number 3376637, and the driver kit, Part Number 3376634.

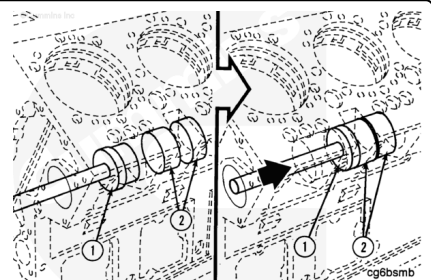
For ease of removal/installation, hydraulic cam bushing tools are available. They include hydraulic actuator kit, Part Number 3823621, installation/removal kit, Part Number 3823647, and centering guide kit, Part Number 3823653.



Use the driver assembly (1). Remove the camshaft bushing (2) from the bore that contains one bushing.



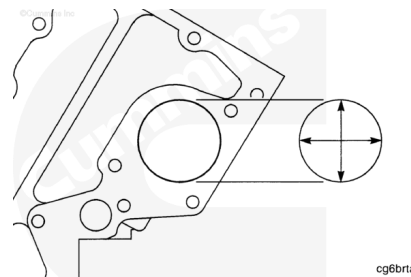
Use the driver assembly (1). Push one of the bushings (2) until it touches the other. Push both bushings (2) out at the same time.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

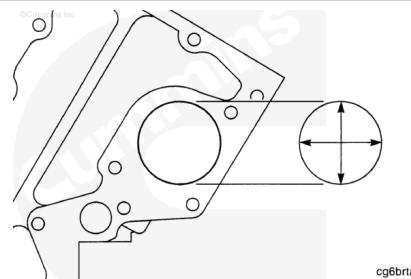
Use an emery cloth. Remove any burrs and sharp edges from the bore.

Use solvent to clean the bore.

Dry with compressed air.

Measure the inside diameter.

Camshaft Bushing Bore in the Block		
mm		in
80.949	MIN	3.1869
81.013	MAX	3.1894



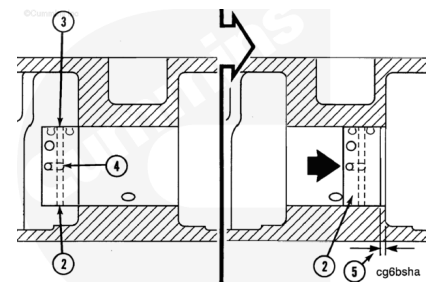
If the inside diameter is **not** within specifications, repair the block by installing a repair sleeve.

See the Alternative Repair Manual, Bulletin 3379035.

Install

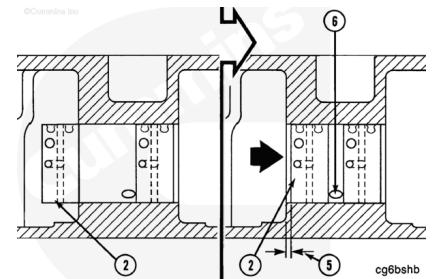
To install the first bushing in the bores that require two bushings:

- Position the bushing (2) so that the clinch joint of the bushing (3) is pointed toward the cylinder head surface of the block. The elongated hole (4) is to be on the side of the bore that is nearest to the vee of the block.
- Use a driver assembly. Push the bushing through the bore. The bushing **must** be even with, or no more than 1.6 mm [0.0625 inch] below the surface of the block web (5).



Position the second bushing (2) for the bore as described previously.

- Use the driver assembly. Push the bushing in the bore until it is even with, or no more than 1.6 mm [0.0625 inch] below the surface of the block web (5).
- Make sure the oil hole (6) is open and clean.



To install the single bushings at the LB front and RB rear:

- Position the bushing (2) so that the elongated hole in the bushing (4) is in line with the oil hole in the block (6). The oil holes in the bushing will be nearest to the cylinder block. The clinch joint (3) will be pointed toward the outboard side of the bore.
- Use the driver assembly. Push the bushing in the bore until it is even with, or no more than 1.6 mm [0.0625 inch] below the surface of the block web (5).
- Check to make sure the oil hole in the block (6) is aligned with the elongated hole in the bushing (4).
- Make sure the oil hole (6) is open and clean.

